

SYDNEY TECHNICAL HIGH SCHOOL



YEAR 11 MATHEMATICS

PRELIMINARY ASSESSMENT TASK 1

MAY 2012

Time Allowed: 70 MINUTES

Instructions

1. Attempt all questions
2. Start each question on a new page
3. Show all necessary working
4. Write your name and your teachers name on each booklet used
5. Approved calculators may be used

QUESTION 1

(10 MARKS)

MARKS

- a) Calculate $\frac{3.7 \times 8.9}{\sqrt{16.94} + 10}$ to 4 significant figures. 2
- b) Write $0.2\dot{6}$ as a fraction in its lowest terms, showing working. 2
- c) Evaluate $\sqrt[4]{256}$ 1
- d) A manufacturer produces an item for \$225. At what price must the item be sold to make a profit of 40% ? 1
- e) Evaluate $|2^2 - 8| - |-7|$ 1
- f) (i) Express $\sqrt[3]{x^5}$ in index notation. 1
(ii) Express $\frac{1}{3x^2}$ using negative indices. 1

QUESTION 2 (9 MARKS)

MARKS

- a) An insect weighs 2.3×10^{-3} grams. How much would 2500 of these insects weigh? 1
- b) Simplify (i) $2x - 4y + 6x - 9y$ 1
(ii) $(3\sqrt{7})^2$ 1
- c) Simplify $2\sqrt{75} + 4\sqrt{147}$ 2
- d) Find m and n if $m - n\sqrt{5} = (3 - \sqrt{5})^2$ 2
- e) Express $\frac{4 + \sqrt{3}}{2\sqrt{6}}$ with a rational denominator. 2

QUESTION 3 (9 MARKS)

MARKS

a) Expand and simplify

$$(i) \ 5(x - 2) + 3(2x - 9) \quad 2$$

$$(ii) \ (x - 1)^2 - (x + 2)(2x + 5) \quad 3$$

b) Factorise

$$(i) \ 2x^2 + 3x - 2 \quad 1$$

$$(ii) \ 8m^3 + 125 \quad 1$$

$$(iii) \ xy - y^2 - x + y \quad 2$$

QUESTION 4 (9 MARKS)

$$a) \quad \text{Simplify} \quad (i) \ \frac{9m + 6}{3m^2 + 2m} \quad 2$$

$$(ii) \ \frac{x^2 - x - 20}{x^2 - 25} \div \frac{x + 1}{x^2 + 5x} \quad 3$$

$$b) \quad \text{Factorise} \ (2x - 3)^2 - 25 \quad 2$$

$$c) \quad \text{Express} \ \frac{2x}{5} - \frac{x + 1}{10} \text{ as a single fraction in its lowest terms.} \quad 2$$

QUESTION 5 (8 MARKS)

MARKS

- | | | |
|-------|-------------------------------------|---|
| Solve | a) $3t - 3 = 5t + 8$ | 2 |
| | b) $ 2x + 3 = 11$ | 2 |
| | c) $\frac{7}{a} + 2 = \frac{3}{2a}$ | 2 |
| | d) $x^2 + 5x = -6$ | 2 |

QUESTION 6 (10 MARKS)

- | | | |
|----|---|---|
| a) | The curved surface area (A) of a cylinder is given by $A = 2\pi rh$.
Find the height (h) if the area of the curved surface is $132\pi \text{ cm}^2$
and the radius is 6cm . | 2 |
| b) | Solve $3x - 4 > x + 7$ and graph its solution on a number line. | 3 |
| c) | Solve simultaneously $2x + y = 8$
$3x + 2y = 13$ | 3 |
| d) | Make y the subject of $x = \sqrt{\frac{A}{y}}$ | 2 |

QUESTION 7 (10 MARKS)

MARKS

- a) Solve $|2x - 5| \geq 7$ 2
- b) Sketch the following on separate diagrams, stating the domain and the range:
- (i) $2x + 3y - 6 = 0$ 3
- (ii) $y = x^2 - 4$ 3
- c) Given that $f(x) = 3x^2 - 5x + 2$, evaluate $f(-2)$. 2

QUESTION 8 (9 MARKS)

MARKS

- a) Simplify $\frac{3}{x^2 + 2x + 1} + \frac{3}{x^2 - 1}$ 3
- b) Solve $4x^2 + 12x + 1 = 0$, leaving your answer in simplest surd form. 3
- c) If $x = (\frac{2}{3})^3$ and $y = (\frac{1}{2})^2$, find the exact value of $x^2 y^4$ 3

QUESTION 1 (9)

a) $2.33284... = 2.333$ (4 sig figs) 2

b) let $x = 0.2666...$

$10x = 2.666...$

$100x = 26.666...$

$90x = 24$

$x = \frac{24}{90} = \frac{4}{15}$

$\therefore 0.2\bar{6} = \frac{4}{15}$

c) 4

d) 140% of \$225 = \$315

e) -3

f) i) $x^{\frac{5}{3}}$

ii) $3^{-1}x^{-2}$

QUESTION 2 (9)

a) $2500 \times 2.3 \times 10^{-3} = 5.75g$

b) i) $8x - 13y$

ii) 63

c) $10\sqrt{3} + 28\sqrt{3} = 38\sqrt{3}$

d) $m - n\sqrt{5} = 9 - 6\sqrt{5} + 5 = 14 - 6\sqrt{5}$

$m = 14$ $n = 6$

e) $\frac{4+\sqrt{3}}{2\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}} = \frac{4\sqrt{6}+\sqrt{18}}{12}$

$= \frac{4\sqrt{6}+3\sqrt{2}}{12}$

QUESTION 3 (9)

a) i) $5x - 10 + 6x - 27 = 11x - 37$

ii) $x^2 - 2x + 1 - (2x^2 + 5x + 4x + 10)$

$= -x^2 - 11x - 9$

b) i) $(2x-1)(x+2)$

ii) $(2m+5)(4m^2-10m+25)$

iii) $y(x-y) - (x-y) = (x-y)(y-1)$

QUESTION 4 (9)

a) i) $\frac{3(3m+2)}{m(3m+2)} = \frac{3}{m}$

ii) $\frac{(x-5)(x+4)}{(x-5)(x+5)} \times \frac{x(x+5)}{x+1} = \frac{x(x+4)}{x+1}$

b) $(2x-3-5)(2x-3+5) = (2x-8)(2x+2)$
 $= 4(x-4)(x+1)$

c) $\frac{4x-x-1}{10} = \frac{3x-1}{10}$

QUESTION 5 (8)

a) $2t-3=5t+8$

$-11 = 3t$

$t = -\frac{11}{3}$

b) $2x+3=11$ $-2x-3=11$

$2x=8$

$-2x=14$

$x=4$

$x=-7$

c) $\frac{7}{a} + 2 = \frac{3}{2a}$

$14 + 4a = 3$

$4a = -11$

$a = -\frac{11}{4}$

d) $x^2+5x+6=0$

$(x+2)(x+3)=0$

$x=-2, x=-3$

QUESTION 6 (10)

a) $A = 2\pi rh$

$132\pi = 2\pi \times 6h$

$h = 11$ $\therefore$ Height is 11cm

b) $3x-4 > x+7$

$2x > 11$

$x > \frac{11}{2}$

c) $2x+y=8$ - ①

$3x+2y=13$ - ②

① $\times 2$: $4x+2y=16$ - ③

③ - ②: $x=3$

Sub in ①: $6+y=8$

$y=2$

$\therefore x=3, y=2$

d) $x = \sqrt{\frac{A}{y}}$
 $x^2 = \frac{A}{y}$
 $y = \frac{A}{x^2}$

QUESTION 7 (10)

a) $|2x-5| \geq 7$

$$2x-5 \geq 7 \quad -2x+5 \geq 7$$

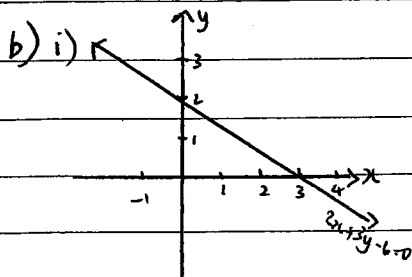
$$2x \geq 12$$

$$-2x \geq 2$$

2

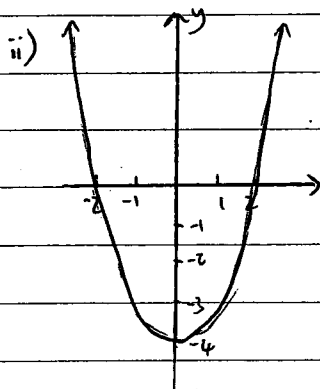
$$x \geq 6$$

$$x \leq -1$$



Domain: All real x values

Range: All real y values



Domain: all real x values

Range: $y \geq -4$

c) $f(x) = 3x^2 - 5x + 2$

$$f(-2) = 3(-2)^2 - 5(-2) + 2$$

$$= 24$$

QUESTION 8 (9)

a) $\frac{3}{(x+1)(x+1)} + \frac{3}{(x-1)(x+1)}$

$$= \frac{3(x-1) + 3(x+1)}{(x+1)(x+1)(x-1)}$$

3

$$= \frac{3x-3+3x+3}{(x+1)^2(x-1)}$$

$$= \frac{6x}{(x+1)^2(x-1)}$$

b) $4x^2 + 12x + 1 = 0$

$$x = \frac{-12 \pm \sqrt{144 - 4 \times 4 \times 1}}{2 \times 4}$$

$$= \frac{-12 \pm \sqrt{128}}{8}$$

3

$$= \frac{-12 \pm 8\sqrt{2}}{8}$$

$$= \frac{-3 \pm 2\sqrt{2}}{4}$$

c) $x^2 y^4 = \left(\left(\frac{2}{3}\right)^2\right)^2 \left(\left(\frac{1}{2}\right)^2\right)^4$

$$= \frac{2^4}{3^4} \times \frac{1}{2^8}$$

3

$$= \frac{1}{2^4 3^4}$$

$$= \frac{1}{1296}$$